$$
\begin{aligned}
s_t^0 &= h_\theta(o_{1:t}) \\
(r_t^k, s_t^k) &= g_\theta(s_t^{k-1}, a_{t+k}) \\
(p_t^k, v_t^k) &= f_\theta(s_t^k)
\end{aligned}
$$